# Notes: Voss (RFS, 2025)

## Short-Term Debt and Corporate Governance

## Contents

# 1 Big picture

- **Question.** How does short-term debt affect shareholder governance through blockholder exit and voice?
- **Standard view.** Short-term creditors improve governance because they roll over frequently and can respond quickly to new information about managerial misbehavior.
- **Main contribution.** Voss shows that this same information sensitivity can undermine governance. Short-term creditors learn from blockholder trades and public interventions; their rollover response feeds back into prices and shareholder payoffs.
- **Baseline mechanism.** A blockholder privately observes bad news and may exit. Her sale depresses the share price. Short-term creditors infer bad fundamentals from the price/order flow and demand higher rollover rates or refuse rollover. Because the higher debt burden reduces shareholder value, the blockholder's exit price is lower.
- **Key comparative static.** Moderate short-term debt can make prices more sensitive to information and strengthen governance. High short-term debt makes exit noncredible because exiting itself triggers an adverse credit-market response.
- **Managerial incentives.** The manager exerts effort only if the threat of blockholder exit is credible and prices are informative. The effort cutoff is hump-shaped in the share of short-term debt.
- **Voice channel.** The paper studies both hidden behind-the-scenes monitoring and public proxy-style campaigns. Short-term debt can initially improve voice incentives, but high short-term debt again reduces them.
- **Main tension.** Short-term debt increases price sensitivity but reduces price informativeness when it deters informed trading by the blockholder.
- **Prediction.** Conditional on blockholder exit, the share price drop should be larger when short-term debt is higher. However, the probability or volume of exit should fall with short-term debt.
- **Corporate governance takeaway.** Firms that rely heavily on short-term funding, such as banks, may face a governance problem: the threat of exit and voice by blockholders becomes weak when creditor rollover reacts too strongly to bad news.

# 2 Model primitives and measurement

## 2.1 Players and timing

- There are three dates, $t = 1, 2, 3$.
- The firm has a manager $M$, a blockholder $B$, dispersed shareholders, liquidity traders, a competitive market maker, and creditors.
- The firm chooses a debt maturity structure with a fraction $\gamma$ of debt issued as short-term debt and $1 - \gamma$ as long-term debt.
- At $t = 1$, financing occurs and the manager chooses effort $a \in \{0, 1\}$.
- At $t = 2$, the blockholder observes the state $S \in \{S_L, S_H\}$ and may exit by selling her block.
- The market maker observes total order flow $Q$ and sets the share price $P(Q)$.
- Short-term creditors observe the price and decide whether to roll over.
- At $t = 3$, payoffs are realized.

**Interpretation:** The timing is designed to create an informational spillover from the equity market to the debt market. Blockholder trades are informative for prices, and prices are informative for short-term creditors.

## 2.2 Debt maturity structure

The fraction of short-term debt is

$$\gamma \in [0, 1]. \tag{1}$$

Long-term debt is fixed until maturity. Short-term debt must be rolled over at $t = 2$, so its face value at rollover depends on creditors' posterior beliefs after observing the price.

**Interpretation:** The paper isolates the effect of short-term creditors' rapid response to information. The debt maturity structure is not just a financing choice; it affects governance through feedback from prices to rollover terms.

## 2.3 Managerial payoff

The manager's payoff is a weighted sum of the interim share price, terminal shareholder value, and effort cost:

$$\omega_p P + \omega_v V_{t=3} - \mathbf{1}_{a=1} c. \tag{1}$$

Here $c$ is the manager's private effort cost, and $\omega_p, \omega_v \geq 0$ measure how much the manager cares about the share price and terminal shareholder value.

**Interpretation:** The manager works only if the governance system makes effort valuable enough. Share price informativeness and blockholder exit therefore affect managerial incentives.

## 2.4 Blockholder exit and order flow

- The blockholder observes the state $S$ privately.
- She sells only after bad news, $S_L$.
- Liquidity traders sell $\phi$ shares or zero shares with equal probability.
- Total order flow is

$$Q \in \{-2\phi, -\phi, 0\}.$$

- The market maker sets $P(Q)$ to break even conditional on order flow.

**Interpretation:** Liquidity trading lets the blockholder camouflage her sale. But when the sale is detected, the order flow reveals adverse information.

## 2.5 Market-maker pricing condition

The share price equals expected shareholder value conditional on order flow:

$$P(Q) = E\left[\max\{R - \gamma D_{ST}^2(P(Q)) - (1-\gamma)D_{LT}, 0\} \mid Q\right]. \tag{2}$$

**Interpretation:** Equation (2) is a fixed point. The price affects short-term debt rollover terms, and rollover terms affect shareholder value, which then determines the price.

### 2.6 Rollover condition

At $t = 2$, short-term creditors roll over if the promised new face value makes them break even:

$$D^1_{ST} = [\pi(Q)p_H + (1 - \pi(Q))p_L]\, D^2_{ST}(Q). \tag{3}$$

The promised rollover face value also cannot exceed what the firm can credibly repay:

$$D^2_{ST}(Q) \leq \frac{R - (1-\gamma)D_{LT}}{\gamma}. \tag{4}$$

**Interpretation:** Bad order flow lowers creditors' beliefs about success and raises the rollover rate. If short-term debt is high enough, rollover after bad news may become impossible.

# 3 Models and empirical specifications

### 3.1 Baseline model: exit as governance

The baseline model studies the threat of exit. The blockholder's exit is valuable for governance because it makes prices informative. However, when short-term debt is high, exiting creates a negative feedback loop:

$$\text{exit} \Rightarrow \text{bad order flow} \Rightarrow \text{lower price} \Rightarrow \text{higher rollover rate} \Rightarrow \text{lower shareholder value.}$$

**Interpretation:** Exit is a disciplinary device only if it is credible. High short-term debt can make exit too costly for the blockholder.

### 3.2 Lemma 1: prices reveal order flow

Fix any equilibrium with positive exit probability. The equilibrium price function $P^*(Q)$ is perfectly informative about total order flow $Q$.

**Interpretation:** Because the market maker must price expected shareholder value, and shareholder value depends on creditors' response to the price, different order flows must induce different prices. Creditors can infer $Q$ from $P(Q)$.

### 3.3 Proposition 1: paralyzing effect of short-term debt

Under the paper's regularity assumption, there is a unique equilibrium exit strategy $\eta^*$ and cutoffs $\underline{\gamma}$ and $\overline{\gamma}$ such that:

1. if $\gamma \leq \underline{\gamma}$, then $\eta^* = 1$;
2. if $\gamma \in (\underline{\gamma}, \overline{\gamma})$, then $\eta^* \in (0, 1)$ and strictly decreases in $\gamma$;
3. if $\gamma \geq \overline{\gamma}$, then $\eta^* = 0$.

**Interpretation:** More short-term debt eventually paralyzes exit. Once the debt rollover response is severe enough, the blockholder does not sell even after bad news.

### 3.4 Exit indifference condition

In the mixed region, the blockholder is indifferent between exiting and retaining her shares:

$$\alpha \left[\frac{1}{2}P(-\phi) + \frac{1}{2}P(-2\phi)\right] = \alpha \left[\frac{1}{2}V(S_L, -\phi) + \frac{1}{2}V(S_L, 0)\right]. \tag{5}$$

**Interpretation:** More short-term debt lowers exit profits and changes the payoff from staying. The equilibrium exit probability must fall enough to maintain indifference.

### 3.5 Managerial incentives and Proposition 2

The manager's cutoff type $\hat{c}$ solves

$$\hat{c} = \omega_p \Delta q\, \eta^* \frac{1}{2}\left[P(0) - P(-2\phi)\right] \tag{2}$$

$$+ \omega_v \Delta q \frac{1}{2}\Big[\eta^* \{V(S_H, -\phi) + V(S_H, 0) - V(S_L, -2\phi) - V(S_L, -\phi)\} \tag{3}$$

$$+ (1 - \eta^*)\{V(S_H, -\phi) + V(S_H, 0) - V(S_L, -\phi) - V(S_L, 0)\}\Big]. \tag{6}$$

Under the paper's assumptions, there is a unique cutoff $\hat{c} \in (0, c)$ such that manager types $c \leq \hat{c}$ work and types $c > \hat{c}$ shirk. The cutoff is hump-shaped in $\gamma$.

**Interpretation:** Moderate short-term debt improves managerial incentives by making prices more sensitive to exit. High short-term debt reduces managerial incentives by making exit less likely and prices less informative.

### 3.6 Hidden voice and Proposition 3

The paper extends the model so that the blockholder can engage in hidden behind-the-scenes monitoring. Monitoring changes the distribution of manager types and costs the blockholder $\kappa$.

Monitoring is an equilibrium if

$$\kappa \leq \bar{\kappa} = [G^m(\hat{c}) - G(\hat{c})]\Delta q\,(V_H(\hat{c}) - V_L(\hat{c}))\,. \tag{9}$$

**Interpretation:** The incentive to monitor depends on the probability that monitoring makes the manager work and on how much more valuable the good state is for the blockholder. This incentive is also hump-shaped in short-term debt.

### 3.7 Public voice and Proposition 4

The paper then studies public campaigns such as proxy contests. Public voice can improve value with probability $\delta$, but it also reveals that the firm may be in the bad state.

Proposition 4 shows that voice incentives are positive only if $\delta > 1/2$ and are hump-shaped in $\gamma$. The voice-incentive maximizing level is interior:

$$\gamma_V^* = \widetilde{\gamma} < 1. \tag{4}$$

**Interpretation:** Public voice becomes risky when short-term creditors interpret the campaign as a signal of bad news. High short-term debt makes public engagement less attractive because failed voice triggers worse refinancing terms.

## 4 Identification and empirical design

### 4.1 Nature of the paper

This is a theoretical paper. Its "identification" is conceptual: it isolates a feedback channel between equity-market governance and short-term credit rollover.

**Interpretation:** The key experiment is a comparative static in $\gamma$, the fraction of short-term debt.

## 4.2 What the model predicts

With exogenous variation in short-term debt, the model predicts:

- conditional on blockholder exit, the share price decline should be larger when short-term debt is higher;
- the probability or volume of exit should decrease in the level of short-term debt;
- share prices should become less informative when short-term debt is high;
- governance through exit, behind-the-scenes intervention, and public campaigns should weaken when short-term debt is high;
- block formation, as measured by 13D/13G filings, should be weaker at high short-term debt levels.

**Interpretation:** The unconditional effect on price volatility can be ambiguous because price reactions become larger conditional on exit, but exit becomes less likely.

## 4.3 What makes the mechanism distinctive

- The harm from short-term debt does not require exogenous liquidation costs.
- The feedback works through endogenous refinancing terms set by short-term creditors.
- The same debt maturity that makes creditors responsive also discourages the blockholder from producing information through exit or voice.
- The model therefore identifies a dark side of information-sensitive debt for corporate governance.

**Interpretation:** The paper complements classic theories in which short-term debt disciplines managers. Here, short-term debt can discipline the wrong party: it deters the informed shareholder from acting.

# 5 Tables, figures, and original equation panels

The paper has no empirical tables. The central visuals are theoretical figures and equation panels. The panels below are directly cropped from the paper and grouped compactly.

## 5.1 Figure 1 and Figure 4: timing of exit and voice models

**Read:**

- Figure 1 gives the timing of the baseline model: financing, managerial action, blockholder exit, market pricing, creditor rollover, and final payoffs.
- Figure 4 adds hidden monitoring by the blockholder before managerial action.
- The key difference is that the blockholder can use voice before prices reveal her information.

**Economic message:** The timing clarifies the central spillover: stock-market trades occur before short-term creditors roll over debt, so prices affect refinancing terms.

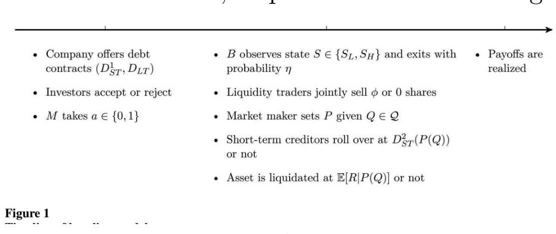


*Figure 1: Baseline timing*

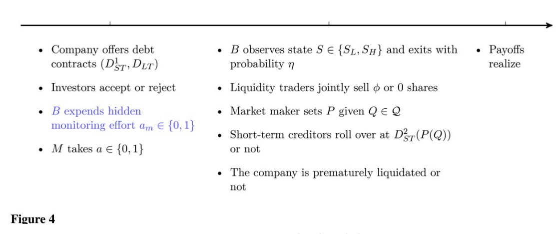


*Figure 4: Timing with hidden voice*

## 5.2 Equation panels: pricing and rollover feedback

**Read:**

- Equation (2) is the market maker's zero-profit pricing condition.
- Equations (3) and (4) give short-term creditors' rollover break-even condition and the credible repayment bound.
- Together they produce a fixed point: prices determine rollover rates, and rollover rates determine shareholder value and prices.

**Economic message:** This is the formal feedback effect. Equity-market information changes credit-market terms, which then change equity value.

rule. The share price is then given by the market maker's zero-profit conditior which demands that the share price equals the expected shareholder value that is,

$$P(Q)=\mathbb{E}[max\{R-\gamma\, D_{ST}^{2}(P(Q))-(1-\gamma)D_{LT};0\}|Q]. \qquad (2$$

*Equation (2): share price fixed point*

At $t=2$, conditional on observing $P(Q)$ and all other short-term credito rolling over, a short-term creditor's break-even condition is

$$D_{ST}^{1}=[\pi(Q)p_H+(1-\pi(Q))p_L]D_{ST}^{2}(Q). \qquad ($$

Note that by Lemma 1, creditors' posterior belief can be directly conditione on $Q$. The left side of Equation (3) is the face value promised to a short-ter creditor in the first period. It has to be equal to the expected repayment the creditor rolls over at face value $D_{ST}^{2}(Q)$, assuming that the company continued. Put differently, the right side is precisely the expected payment a outside investor would need to be promised by the company to be willing

*Equations (3)-(4): rollover terms*

## 5.3 Figure 2 and Equation (5): blockholder exit

**Read:**

- Figure 2 shows that the equilibrium exit probability is weakly decreasing in short-term debt.
- Exit is certain at low $\gamma$, mixed at intermediate $\gamma$, and absent at high $\gamma$.
- Equation (5) gives the blockholder's indifference condition in the mixed region.

**Economic message:** High short-term debt makes exit noncredible. The blockholder internalizes part of the negative refinancing response her trade causes.

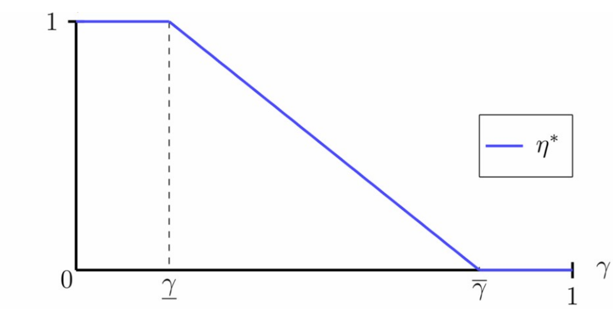


*Figure 2: Exit probability*

in equilibrium for $\gamma\in(\underline{\gamma},\overline{\gamma})$. Her indifference condition at $t=2$ requires that

$$\underbrace{\alpha\left(\frac{1}{2}P(-\phi)+\frac{1}{2}P(-2\phi)\right)}_{=\Pi^E}=\underbrace{\alpha\left(\frac{1}{2}V(S_L,-\phi)+\frac{1}{2}V(S_L,0)\right)}_{=\Pi^{NE}(\eta^*)}. \qquad (5)$$

More short-term debt decreases exit profits $\Pi^E$ but, for a fixed exit likelihood $\eta^*$, increases retention profits $\Pi^{NE}(\eta^*)$. Intuitively, more information sensitive

*Equation (5): exit indifference*

## 5.4 Figure 3 and Equation (6): managerial incentives

**Read:**

- Figure 3 shows managerial incentives as a hump-shaped function of short-term debt.
- Equation (6) defines the manager's effort cutoff.
- Low-to-moderate short-term debt makes prices more sensitive to managerial action.
- High short-term debt lowers the probability of exit and thereby reduces price informativeness.

**Economic message:** Governance is strongest at an interior maturity structure. Neither all long-term debt nor very high short-term debt maximizes managerial effort.

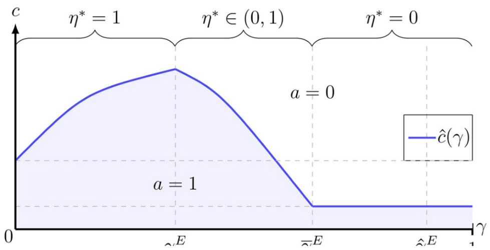


*Figure 3: Managerial incentives*

incentives. $M$'s payoff (1) from taking the firm value-increasing action $a=1$ strictly decreases in his true type $c$. Hence, in any equilibrium, all manager types below some threshold $\hat{c} \in (0,\overline{c})$ work and all manager types above it shirk. At $\hat{c}$, $M$ is indifferent between working and shirking where $\hat{c}$ is the solution to

$$\hat{c} = \omega_p \Delta_q \eta^* \frac{1}{2}[P(0) - P(-2\phi)]$$

$$+\omega_v \Delta_q \frac{1}{2}[\eta^*(V(S_H,-\phi)+V(S_H,0)-V(S_L,-2\phi)-V(S_L,-\phi))$$

*Equation (6): effort cutoff*

## 5.5 Figure 5 and hidden voice incentives

**Read:**

- Figure 5 shows that private voice incentives are also hump-shaped in short-term debt.
- The blue/red payoff components widen at low debt maturity levels and narrow at intermediate levels.
- Equation (9) expresses monitoring incentives as the product of monitoring's effect on effort probability and the blockholder's payoff gain from the good state.

**Economic message:** Exit and voice are complements. If short-term debt destroys exit credibility, it also weakens the incentives for behind-the-scenes monitoring.

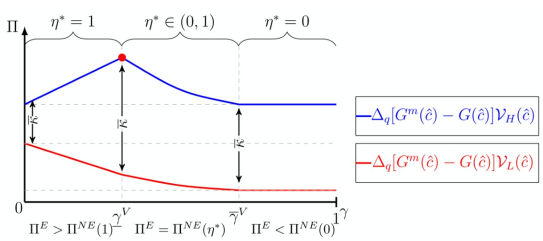


*Figure 5: Voice incentives*

Fix an equilibrium in which $B$ monitors, along with the associated equilibrium conjectures of $\hat{c}(a_m^*=1)$ and $\eta^*(a_m^*=1)$. For brevity, I write $\hat{c}$ and $\eta^*$ henceforth. Then, $B$'s expected payoff is

$$(q+G^m(\hat{c})\Delta_q)\,\mathcal{V}_H+(1-q-G^m(\hat{c})\Delta_q)\,\mathcal{V}_L-\kappa, \tag{7}$$

where $\mathcal{V}_H := \frac{\alpha}{2}[V(S_H,-\phi)+V(S_H,0)]$ is $B$'s payoff conditional on $S_H$, and $\mathcal{V}_L := max\{\Pi^E;\Pi^{NE}(\eta^*)\}$ represents the maximal payoff $B$ can obtain from exit or share retention in state $S_L$. $B$'s expected profits from deviating to $a_m=0$ are

$$(q+G(\hat{c})\Delta_q)\,\mathcal{V}_H+(1-q-G^m(\hat{c})\Delta_q)\,\mathcal{V}_L. \tag{8}$$

Importantly, a deviation to not monitoring does not change the cutoff $\hat{c}$ because monitoring effort is hidden from outsiders. Consequently, interest rates and the share price are not affected directly by $B$'s deviation to $a_m=0$. The deviation merely reduces the probability that $M$'s type is below the fixed cutoff $\hat{c}$ from $G^m(\hat{c})$ to $G(\hat{c})$.

**Proposition 3.** Suppose Assumptions 2 and 3 hold. Then, there is an equilib-

*Equations (7)-(9): private voice*

## 5.6 Public voice and short-selling extensions

**Read:**

- Equation (10) summarizes public voice incentives.
- Public voice is beneficial only if the campaign succeeds with probability greater than one half.
- Equations (11)-(13) show that short-selling can be profitable, but the blockholder's existing stake makes her internalize the negative interest-rate feedback.

**Economic message:** The governance problem is not limited to literal share sales. Public activism and short selling are also constrained by the rollover feedback of short-term debt.

trading, I assume $\gamma = 1$ and abstract from management so that $S_H$ realizes with probability $q$. Liquidity traders again sell an amount of $\phi > 0$ shares or do not trade, with equal probability. If $B$ exits, it is easy to show that she optimally (short) sells $\phi$ shares. Conjecture an equilibrium in which $B$ does not sell any shares due to the high level of short-term debt. Consistent with the $D1$ criterion, off-path beliefs are $\pi(-2\phi) = 0$. Then, $B$'s equilibrium profits from share retention are given by

$$\alpha \frac{1}{2}[V(S_L, -\phi) + V(S_L, 0)]. \tag{11}$$

By contrast, if $\phi > \alpha$, selling $\alpha$ shares and shorting the remaining $\phi - \alpha$ shares yields an expected payoff of

$$\begin{aligned} &\alpha \frac{1}{2}[P(-2\phi) + P(-\phi)] \\ &\quad + (\phi - \alpha)\frac{1}{2}[P(-2\phi) + P(-\phi) - V(S_L, -2\phi) - V(S_L, -\phi)] \\ &= \alpha \frac{1}{2}[V(S_L, -2\phi) + V(S_L, -\phi)] \\ &\quad + \phi \frac{1}{2}[P(-2\phi) + P(-\phi) - V(S_L, -2\phi) - V(S_L, -\phi)]. \end{aligned} \tag{12}$$

*Equations (11)-(13): short selling*

if and only if $\mathcal{V}_L(a_{pub} = 1) - \kappa \geq max\{\mathcal{V}_L(a_{pub} = 0), \mathcal{P}\}$. In the Appendix, I show that there exists a unique cutoff $\tilde{\gamma} \leq \hat{\gamma} < 1$ such that for lower levels of short-term debt, the relevant deviation is to exit ($max\{\mathcal{V}_L(a_{pub} = 0), \mathcal{P}\} = \mathcal{P}$), whereas for larger levels it is to stay with the firm ($max\{\mathcal{V}_L(a_{pub} = 0), \mathcal{P}\} = \mathcal{V}_L(a_{pub} = 0)$). Solving for $\kappa$ yields the following proposition.

**Proposition 4.** Suppose Assumptions 2 and 3 hold, and that $\delta \in (\frac{1}{2}, 1)$. Then, public voice ($a_{pub} = 1$) is an equilibrium if

*Equation (10): public voice*

# 6 Short summary

- Short-term debt allows creditors to react quickly to new information.
- This responsiveness can improve governance at moderate levels of short-term debt.
- But when short-term debt is high, blockholder exit reveals bad news and triggers worse rollover terms.
- The adverse rollover response lowers shareholder value and the price the blockholder receives when exiting.
- As a result, the blockholder exits less often or not at all.
- Less exit makes prices less informative and weakens managerial incentives.
- Hidden monitoring and public campaigns are also hump-shaped in short-term debt.
- The model predicts that high short-term debt weakens governance, especially in firms for which refinancing is fragile.
- The main lesson is that information-sensitive debt can undermine information-producing governance by shareholders.

# 7 Conclusion

## 7.1 Core conclusion

Voss shows that short-term debt has a dark side for corporate governance. While short-term creditors can respond quickly to information, their response changes the payoffs from blockholder exit and voice. High levels of short-term debt can therefore make shareholder governance ineffective.

## 7.2 Economic intuition

Blockholder exit is useful because it makes prices informative and disciplines managers. But if the firm relies heavily on short-term funding, the same informative price decline causes creditors to demand worse rollover terms. Since shareholders bear the higher debt burden, the blockholder's sale depresses her own exit price. At high levels of short-term debt, she no longer exits, so managers face weaker discipline.

## 7.3 Takeaway

The paper explains why the governance of short-term-funded firms, especially financial institutions, can be difficult. The very feature that makes short-term debt disciplining - creditor responsiveness to information -

can deter informed shareholders from producing and revealing information in the first place.